

CREATORZZZ

TRL3 Technical Validation Report

SIH26043 • Government of Jharkhand • 10 September 2026

Decision: SIMPLIFY. Use the competent B0 workflow for qualification/readiness review and pilot evidence plans. Retain P2 as an isolated integrity experiment for a timed operator study. The evidence does not justify full custom MVP expansion yet.

The proof is executable. Its deterministic checks largely conform to the frozen synthetic references. The main P1/P3 snapshot results tie the spreadsheet baseline, and the custom P1 returns two additional blockers. No independent human gold, real-case validation, human timing or plain-LLM comparison was obtained. The frozen product purpose is unchanged. [DOC01 §16]

Mechanism	Full result	Observed evidence	Missing gate
P1: qualification + readiness	PARTIAL	44/44 gaps; 0/60 false ready; 10/10 ready recall	Independent gold, real cases, grounding and time advantage
P2: commitments + invalidation	PARTIAL	72/72 affected decisions; 8/8 consistent replays	Independent oracle, operator comparison and ownership
P3: evidence + outcomes	PARTIAL	13/13 defects; 0/13 adequate false flags	Human review, real packets, time advantage and source entailment

Evidence maturity. All 220 input packets/scenarios are SYNTHETIC: 120 P1 cases, 40 P2 scenarios and 60 P3 packets. P2 scenarios contain 120 mutations. “Expected” means the author's frozen construction reference; it does not mean independent ground truth. Human review actions and authority records in fixtures are simulated.

What ran. The 49-test automated suite passed. S processed the full synthetic corpus; primary P1/P3 scores use the family holdouts. The B0 workbook calculated the same pre-reviewed P1/P3 snapshots. P2's separately expressed allocation oracle agreed on 192/192 subsets across 12 small problems.

What did not run. B1 has no configured model endpoint. No stakeholder was contacted; no human timing, field pilot, government integration, production authorization or impact assessment occurred. Targets have not been substituted for achieved results.

Evidence: 06_Benchmark/final/metrics.json, 07_Evidence/unit_final.log, 04_P2/allocation_oracle_results.json, 06_Benchmark/B1_run_status.json.

1. Specification and controlled comparison

The official PS governs requested scope. DOC00 v2.0 freezes product direction; DOC01 v1.0 supplies the mechanism and testing specification. Both uploaded FINAL files contained the same DOC00. DOC01 was separately located and read completely. The Team Explainer was also read, including its diagrams. Source hashes and extracted text are included. No broad product research or new presentation was undertaken.

P1 tests the route and readiness consequences of reviewed evidence. It does not establish whether an unreviewed citizen report is true. **P2 tests** exact-version commitments, reservations, changes and affected decisions. **P3 tests** structural evidence sufficiency and conservative outcome recording. [DOC01 §§4-8]

Corpus	Frozen size and split	Evidence shortfall
P1	120 cases; 30 dev / 30 tuning / 60 holdout; 24 context families	0 real cases; target at least 30
P2	40 scenarios / 120 mutations; 20 contexts; 12 allocation problems	No independent human oracle or timed operator sessions
P3	60 packets; 20 dev / 40 holdout; 12 context families; 20 adequate / 20 defective / 20 ambiguous overall	0 real packets; target at least 20

Dataset version **synthetic-v1.0** was frozen before implementation. Release **trl3-0.1-frozen** froze the policy, prompts, thresholds, source code, tests and evaluator before the final run. The tuning partition was not used to change rules. Final outputs, development differences and hashes are preserved. [DOC01 §12]

Cases cover routine grievances, public-value limits, systemic issues, uncertainty, unknown ownership, parallel service/research, unavailable faculty, available lower-ranked capacity, revoked/expired commitments, scope changes, retries, baseline/denominator failures and outcome uncertainty. Domains are low-risk administrative processes in fictional locations; no clinical or dangerous intervention is used.

Limit of the holdout. Family splits separate fictional contexts, while failure archetypes recur across splits. The same assistant authored the fixtures and implementation. This is a frozen conformance test, not an independently authored blind generalisation benchmark.

Gold discipline. System inputs omit expected answers. Construction expectations live in `reviewer_only`; all independent reviewer/adjudicator fields remain null. The runner supplies S only code, policy and input JSON in a separate process, then the evaluator reads references. This is input separation, not an adversarial filesystem security boundary.

2. P1 — qualification and readiness

The executable P1 consumes reviewed Q1–Q9 answers, evidence records, conditions, commitments and sign-offs. It checks relevant source type/scope, role, expiry, exact commitment version and interval coverage. It preserves SATISFIED, UNSATISFIED, UNKNOWN, DISPUTED and EXPIRED. [DOC01 §§4–5]

Unresolved mandatory conditions produce BLOCKED. Current complete conditions produce REVIEW_READY. Only a supplied authorized fixture-human sign-off permits PILOT_READY. A changed dependency of an existing active decision produces REVIEW_REQUIRED and preserves the blocker list. The route array supports SERVICE + RESEARCH_REVIEW, resolving DOC01's single-enum example against its explicit parallel-route table.

Holdout metric	B0 snapshot	S
Critical-gap recall	44/44	44/44
False-ready / all cases	0/60	0/60
False-ready / reference nonready cases	0/50	0/50
Ready-case recall	10/10	10/10
Qualification route-set agreement	60/60	60/60
Blocker precision	44/44	44/46
Unresolvable returned evidence IDs	Not scored	0/552

Preserved differences: PQ066 and PQ096. The reference identifies an unknown-owner blocker. S also blocks accepted scope because its eligibility guard requires an innovation-qualified challenge. Both systems return RESEARCH_REVIEW and BLOCKED. The extra flags may add unnecessary work. B0 performs better on this narrow blocker-precision comparison. Independent policy adjudication must decide whether the extra scope dependency is justified. The development analogue PQ006 is preserved too.

The adversarial cases include an irrelevant official-looking attachment, a reviewed handwritten-looking note, prompt injection, missing permission, missing owner, Hinglish ambiguity and an AI-origin sign-off. These test respect for supplied review records and authority boundaries; they do not benchmark autonomous authenticity detection or language understanding.

An always-abstain negative control has 0/60 false ready but 0/10 ready recall, failing the recall gate. Thus zero false ready alone cannot pass the test. The control is explicitly an ablation, not B0.

Result: PARTIAL. The numerical snapshot gates meet DOC01's zero false-ready, at least 95% gap recall and at least 80% ready recall targets. The required at least 20% reduction in median active human review time is unmeasured. Source entailment, human overrides and independent gold remain pending. A valid reference ID is not proof that a claim is grounded. [DOC01 §12.2]

3. P2 — commitment and dependency integrity

The local SQLite proof implements PROPOSED, OFFERED, ACCEPTED, DECLINED, WITHDRAWN and EXPIRED. Each commitment carries actor, scope, effort/resource, dates, constraints, version, expiry and accepting actor. Accepted scope/term changes create a new OFFERED version requiring reacceptance. [DOC01 §§6–7]

Transactions combine reservation checking with the write. Resources use explicit aliases and UTC half-open intervals, permitting adjacent slots. Expected-version checks reject stale updates. Exact duplicate idempotency keys return their saved result; conflicting reuse is rejected. A history table preserves earlier versions and rejects ordinary update/delete attempts.

Registered suite metric	Observed numerator / denominator
Mutation response agreement	120/120 mutations
Critical invalidation recall	72/72 expected newly affected decisions
Affected-decision precision	72/72 newly reported affected decisions
Snapshots containing double booking	0/120 post-event snapshots
Successful unauthorized actor attempts	0/4 deliberately unauthorized requests
Rejected mutations with unchanged history	48/48 rejected requests
Identical replay result and database export	8/8 duplicate deliveries
Unrelated/closed decisions unchanged	240/240 control-decision observations
Allocation oracle agreement	192/192 subsets across 12 problems

Actual trace, CM001. Faculty commitment v3 is withdrawn, creating v4. CM001-READY and its dependent CM001-OUTCOME reopen as REVIEW_REQUIRED. The unrelated CM001-OTHER and inactive CM001-CLOSED remain unchanged. A duplicate delivery returns the same business result without another mutation. A new-key stale v3 acceptance is rejected as VERSION_CONFLICT.

Replays are excluded from new-invalidation recall and precision denominators. The 240 control observations are repeated per-event states, not 240 independent cases. The allocation oracle uses integer slot occupancy, separately from the engine's interval logic; its authorship is not independent human review. A unit test also exercises two concurrent connections accepting the same exclusive slot: one succeeds and one is rejected. This is not a load test.

Result: PARTIAL. Registered local integrity checks pass. No missed or extra invalidations were observed in this suite. The at least 12 human-reviewed scenarios and at least 20% active-time improvement over B0 are absent. Clock-event delivery, actual acceptance authority, external resource identity and production crash/recovery behavior remain unvalidated. [DOC01 §12.3]

4. P3 — evidence plans and outcome integrity

The structured plan includes outcome, baseline/source/date, metric, unit, denominator, target, method, comparison, time window, evidence owner, reviewer, adverse effects, stop rule, limitations and publication permission. Observations retain their own values, denominator, sample size, dates and references. [DOC01 §8]

The validator checks typed finite values, missing fields, inconsistent units/denominators, short periods, photo-only records and reference presence. It does not infer causal validity from an outcome improvement or an attachment. Signed outcome states are VALIDATED, ITERATE, STOPPED or INCONCLUSIVE; no automatic success narrative or verified causal claim is generated.

Holdout metric	B0 snapshot	S
Critical-defect recall	13/13	13/13
Adequate packets with false critical flags	0/13	0/13
Wrong unambiguous outcome	0/26	0/26
Ambiguous packets kept INCONCLUSIVE	14/14	14/14
Automatic verified causal claims emitted	0/40	0/40
Complete references, all recorded numeric values	Not scored by adapter	195/198
Complete references, VALIDATED numeric values	Not scored by adapter	25/25

The holdout contains 13 adequate, 13 defective and 14 ambiguous packets. S records 5 VALIDATED, 2 ITERATE, 6 STOPPED and 27 INCONCLUSIVE. UNKNOWN/REVIEWER_DISAGREEMENT reference labels are excluded from outcome misclassification and reported separately as conservative abstention coverage.

Preserved missing provenance, PE026. Its observation, denominator and sample size point to a missing source. All three incomplete references remain visible and the packet is INCONCLUSIVE. Do not claim universal quantitative provenance completeness: only 195/198 recorded values have complete references. All 25 numeric fields in the five VALIDATED packets have references, but numeric entailment and authenticity have not been independently checked.

Policy defect to resolve. A stopped pilot with incomplete measurement evidence currently becomes INCONCLUSIVE in the evaluation validator. Actual cessation must remain effective even when evaluation evidence is incomplete. Track operational stop separately from evidence sufficiency before deployment. This issue was documented; the frozen implementation was not retuned.

Result: PARTIAL. The structured defect/false-positive gates meet the at least 90% and at most 10% targets. The real-case, human-adjudication and timing gates do not. DOC01 explicitly permits retaining a manual evidence template when custom automation does not beat it. [DOC01 §§12.4,16]

5. Fair baseline and missing comparisons

B0 is usable. It includes the intake form schema, native spreadsheet case tracker, qualification/readiness checklist, evidence plan, commitments, evidence register and tasks. The workbook has seven sheets. Source references, owners, faculty, acceptance, permission, baseline, metric, reviewer, pilot status and outcome status are included. Companion CSVs support an ordinary task tracker and linked-decision review.

B0 operators receive the same evidence and policy. They can filter, search, use formulas, check calendars, set reminders and reopen affected records manually. No artificial waiting, omitted fields or disabled spreadsheet calculations were used to favor S. The supplied P2 operator aids have no performance score because no operator sessions occurred.

Measured scope is a snapshot. Both B0 and S receive pre-reviewed structured fixture fields. The clerical B0 exporter makes source/role/date predicates explicit. These fields stand in for work that a person must do. Its native formulas match the main P1/P3 results; that does not prove whole-workflow equivalence, and S has not proved that it saves the preprocessing work.

Comparison dimension	Status	Consequence
B0 vs S, P1/P3 structured snapshot	Executed	Main checks tie; B0 has fewer extra P1 flags
B0 vs S, real operator workflow	NOT_MEASURED	No claim of speed or workload advantage
B0 vs S, P2 live change handling	NOT_MEASURED	Integrity engine's operational value remains a hypothesis
B1 plain LLM	NOT_RUN	No model outputs, errors, hallucinations or abstentions to score
Human gold and source entailment	NOT_MEASURED	Construction-reference conformance only
Real social outcome / deployment	NOT_MEASURED	No field-impact or adoption claim

B1 preparation. Fixed prompt B1-plain-v1, explicit model/config fields, raw-output preservation, mechanical scoring adapter and error handling are supplied. No inference endpoint was configured, so model, usage, latency and cost remain null. Individual outputs were neither generated by the current assistant as a substitute nor repaired by hand. A future B1 comparison needs a new preregistered release with an actual model identity.

S wall times for entire local batches, including interpreter startup: 60.325 ms / 120 P1 inputs; 115.667 ms / 40 P2 scenarios; 48.040 ms / 60 P3 inputs. These are single-run execution diagnostics. They are not active reviewer time, a service-latency guarantee or an AI-cost saving.

6. Implementation and authority boundary

The smallest proof is Python plus SQLite, with no external dependencies for core execution. A JSON-lines CLI exposes each mechanism. The benchmark runs S with only source, policy and case inputs, then compares output to the separately stored construction references. The complete code and automated tests are supplied.

Component	Implemented behavior	Material limit
P1 policy evaluator	Condition states, reviewed routes, source/version checks, human sign-off gate	No raw-document interpretation or evidence authenticity
P2 transactional store	Versioned commitments, reservations, replay receipts, dependency traversal, preserved history	One local database; trusted identities/clock; no external scheduler or federation
P3 evidence validator	Typed completeness/consistency checks and conservative signed outcome recording	No causal estimation, source entailment or independent evaluator
Test runner	Frozen inputs/rules, isolated input supply, exact denominators, preserved differences	Same author; no independent blind generalisation
B0 workbook	Ordinary formulas, review fields, tasks and evidence links	Human upkeep and operator error rates unmeasured

Humans must retain evidence verification, systemic need and public-value judgment, owner legitimacy, qualification approval, scoped commitment acceptance, field permission, policy exceptions, readiness sign-off and outcome interpretation. Rejection or withdrawal must remain possible. Software cannot create a resource offer, authority or public-value rationale from missing data. [DOC00; DOC01 §§3–9]

Deterministic logic handles states, required conditions, explicit references, timestamp expiry, current versions, interval conflicts, duplicate keys, dependency closure, finite numeric types and missing fields. AI may later structure text or draft sourced suggestions, but none is necessary for this integrity proof and none was used in S.

Production boundary. HUMAN_FIXTURE and actor-role registries are trusted test inputs, not authentication or legal signatures. The CLI must not be exposed as a public readiness-approval API. A production service needs authenticated server-side identities, decision signatures bound to exact versions, domain policies, actual permission verification and operational audit controls. The local history triggers are not cryptographic tamper-proofing.

Open policy questions include the real owner of readiness, qualifying authority, acceptable evidence by domain, conditional requirements, expiry/waivers, escalation, faculty capacity maintenance, publication/retention and separate operational-stop semantics. No government rule or stakeholder commitment has been invented to fill these gaps.

7. Decision and next execution gate

SIMPLIFY is the single next-build decision. It reflects limited measured advantage and missing required evidence; it is not proof that the whole idea has failed. Retain the frozen problem-qualification and pilot-learning purpose. Reduce custom expansion to what can demonstrate benefit. [DOC01 §16]

Use now: the competent B0 intake, qualification/readiness checklist and evidence-plan template for permissioned case work. **Keep experimental:** P2's commitment/version/dependency kernel. **Defer:** custom P1/P3 expansion, new AI features, the full website and production readiness automation. Preserve the first test run and all disagreements.

Priority	Minimum validation	Decision enabled
1	Sponsor + local authority name the readiness/permission/outcome decision owners and escalation	If no legitimate owner exists, pause custom expansion
2	Two independent gold reviewers; third adjudicator; no tool/reference-answer exposure	Human labels and explicit disagreement record
3	Permissioned at least 30 real/anonymised P1 cases and at least 20 P3 packets	Fresh representative family holdouts
4	Faculty/resource-custodian walkthrough of acceptance, expiry, withdrawal and rework	Valid commitment maintenance and actor rules
5	Counterbalanced B0/S study; at least 12 P2 human scenarios; actual active-time logging	Whether P2 earns at least 20% median time improvement without worse critical errors
6	Pin actual B1 model/version/settings in a new release; preserve every raw output	Plain-LLM comparison, measured errors/usage/cost
7	Adjudicate extra scope blockers, quantitative provenance and stop semantics	Policy revision with a new holdout; no retroactive tuning

Start with at least two relevant study operators for a feasibility study, distinct from gold adjudicators where possible. This is not a power calculation; expand the sample if variability or learning effects prevent a reliable estimate. Rotate arm order, use matched but distinct family cases, record training/assistance and retain incomplete tasks. Do not compare Python execution milliseconds with a person's minutes.

Suggested 14 work days: 1–2 sponsor/rubric review; 3–5 permissioned case collection; 6–8 independent labeling; 9 new release/prompt freeze; 10–12 counterbalanced operator and B1 runs; 13–14 error adjudication and decision. These are planned work days, not booked appointments or completed commitments.

GO requires the complete DOC01 P1/P2/P3 criteria and confirmed operating ownership. A missing human timing value cannot pass the at least 20% reduction gate. No benchmark number here should be presented as measured social impact or a national/generalisation accuracy claim.

8. Reproduction and evidence trail

Extract the package. Core execution requires Python 3.10+ and its standard library. No API credential or package installation is needed for S. From the package directory:

```
PYTHONPATH=src python3 -m unittest discover -s tests -v

python3 scripts/benchmark.py --phase final --output 06_Benchmark/reproduction_01

python3 scripts/allocation_oracle.py
```

The benchmark verifies frozen hashes and refuses to overwrite an existing metrics file. Use a fresh output directory for a labeled reproduction. Do not rerun fixture-authoring scripts over frozen data. The stored B0 formula output supports core replay; the native workbook remains available for inspection and operator use.

Directory	Finished contents
01_B0_Baseline	Form, seven-sheet workbook, trackers/CSVs, checklists, evidence template and operating instructions
02_Test_Fixtures	120 P1 / 60 P3 / 40 P2 inputs, family definitions, separate references/gold workspace, rubric, provenance and freeze
03_P1	Implementation description/schema, actual outputs, test results and preserved failures
04_P2	State machine/schema, 120 mutation traces, results, allocation oracle and limitations
05_P3	Evidence plan/schema, outcome outputs, results and failure analysis
06_Benchmark	B0/S comparison, denominators, frozen evaluator protocol, B1 status/prompt, human plan and empty results template
07_Evidence	Source hashes/text, code freeze, actual logs, screenshot, selected traces, claim and contradiction registers
08_Final_Decision	SIMPLIFY decision, component results, next scope and human validation gates
09_TRL3_Report	This PDF and editable Markdown report

The full executable source, tests, evaluation scripts and README are also included. The screenshot is an actual workbook render; no product interface, field implementation or stakeholder signature is implied.

Source hierarchy and locators. Official PS pp. 1–2 governs the request. DOC00, Final Product & Decision Source of Truth v2.0, fixes direction. DOC01, System Specification + TRL3 Evaluation Protocol v1.0, §§4–8 defines P1/P2/P3; §§12.2–12.4 supplies numerical/data gates; §16 defines Go/Simplify/Pivot. Team Explainer v1.0 supplies team-facing interpretation. Full SHA-256 hashes and extracted paragraphs are in 07_Evidence/source_manifest.json and source_*.txt.

Report status: finished local technical proof and evidence package; independent comparative validation remains incomplete. Recorded results are traceable to the supplied logs and metrics, not invented interviews, labels or benchmark runs.